

MAT 344 HW10

Aditya Pahuja

Due 05/05

Problem 1 (p.227#2). Show that the functions z^n for nonnegative integer n form a normal family in the regions $|z| < 1$ and $|z| > 1$, but not in any region containing a point on the unit circle.

Solution. For every point z in the unit disk, $|z^n| \leq 1$, so by Montel's theorem $\mathcal{F} = \{z^n : n \in \mathbb{Z}_{\geq 0}\}$ is normal on the open unit disk Δ .

To show normality on $\Omega = \{z : |z| > 1\}$, let $\{z^{n_k}\} \subseteq \mathcal{F}$ be a sequence. If $\{n_k\}$ is bounded, then some integer m appears infinitely often among the n_k , so that there is a subsequence $z^m, z^m, \dots$ which trivially converges uniformly to z^m . Otherwise, if $\{n_k\}$ is unbounded, replace $\{n_k\}$ with a subsequence $\{m_k\}$ which is strictly increasing. When $|z| \geq 1 + \varepsilon$ for some fixed $\varepsilon > 0$, we have that for any given $M > 0$, if $k > \log_{1+\varepsilon}(M)$, then

$$|z^{m_k}| \geq (1 + \varepsilon)^k > M.$$

That is, $\{z^{m_k}\}$ converges uniformly to ∞ on the compact annulus $1 + \frac{1}{n} \leq |z| \leq n$ for each positive integer n . These annuli exhaust $\{z : |z| > 1\}$, so we have found a subsequence of $\{z^{n_k}\}$ which converges locally uniformly, proving that $\mathcal{F}$ is normal on this region.

Finally, suppose that Ω is a domain containing a point a on the unit circle. Let $U \subseteq \Omega$ be a neighborhood of a . The sequence $z, z^2, z^3, \dots$ (and any subsequence thereof) converges pointwise to 0 on $U \cap \Delta$, 1 on $U \cap \partial\Delta$, and ∞ on $U - \overline{\Delta}$; this pointwise limit is not continuous, so the convergence cannot have been uniform, meaning $\mathcal{F}$ is not normal on Ω . $\square$

Problem 2 (p.227#3). If $f(z)$ is entire, show that the family of functions $\mathcal{F} = \{f(kz) : k \in \mathbb{C}\}$ is normal in the annulus $r_1 < |z| < r_2$ if and only if f is a polynomial.

Solution. Suppose f is a polynomial. If f is constant then $\mathcal{F}$ is obviously normal, so suppose otherwise and let $\{f(k_n z)\}_{n \in \mathbb{N}}$ be a sequence in $\mathcal{F}$. If $\{k_n\}$ is unbounded, then pick a subsequence $\{k_{n_j}\}$ for which $|k_{n_j}| \rightarrow \infty$; then $\lim_{j \rightarrow \infty} f(k_{n_j} z) = \infty$ for each z in the annulus. This convergence is uniform on compact subsets of the annulus because for a polynomial $p(z) = a_n z^n + \dots + a_1 z + a_0$, $\frac{p(z)}{z^n}$ is arbitrarily close to a_n as z grows arbitrarily large, so that for any $\varepsilon > 0$ there is an $R > 0$ such that if $|z| > R$ then $|p(z)|$ is bounded below by $(|a_n| - \varepsilon)|z^n|$. If on the other hand k_n is a bounded sequence, then

$$\sup_{\substack{n \in \mathbb{N} \\ r_1 < |z| < r_2}} |f(k_n z)| < \infty,$$

which implies the existence of a locally uniformly convergent subsequence on the annulus by Montel's theorem.

Conversely, suppose $\mathcal{F}$ is normal on the annulus and consider the sequence $\{f(nz)\}_{n \in \mathbb{N}}$. Let $s_1, s_2 \in \mathbb{R}$ such that $r_1 < s_1 < s_2 < r_2$; then a subsequence of $f(nz)$ converges uniformly on $s_1 \leq |z| \leq s_2$. If $\{f(nz)\}_{n \in \mathbb{N}}$ is a uniformly bounded sequence, then f is a bounded entire function, so it is constant. If this sequence is unbounded, by applying normality to some subsequence for which $\sup_{r_1 < |z| < r_2} |f(nz)|$ has limit ∞ shows that this subsequence has a further subsequence which converges locally uniformly to ∞ . This implies that f has a pole at ∞ , which for entire functions is equivalent to being a polynomial. $\square$

Problem 3 (p.227#4). If a family $\mathcal{F}$ of holomorphic (or meromorphic) functions is not normal in Ω , show that there exists a point z_0 such that $\mathcal{F}$ is not normal in any neighborhood of z_0 .

Solution. Suppose that for each $z \in \Omega$ there is a neighborhood $U_z \subseteq \Omega$ of z in which $\mathcal{F}$ is normal. Let $K \subseteq \Omega$ be compact and $\{f_n\} \subseteq \mathcal{F}$ a sequence. There are finitely many points $z_1, \dots, z_m$ such that the U_{z_m} cover K . Let $\{f_{n_{1,k}}\}_{k \in \mathbb{N}}$ be a subsequence of $\{f_n\}$ which converges locally uniformly on U_{z_1} , and, for $1 < i \leq m$, define $\{f_{n_{i,k}}\}_{k \in \mathbb{N}}$ to be a subsequence of $\{f_{n_{i-1,k}}\}_{k \in \mathbb{N}}$ which converges locally uniformly on U_{z_i} , which we can do by the normality assumptions. That is, for each K we can find a subsequence $\{f_n^K\}_{n \in \mathbb{N}}$ of $\{f_n\}$ which converges uniformly on K .

Now let $K_1 \subseteq K_2 \subseteq \dots \subseteq \Omega$ be an exhaustion of Ω by compact sets K_j and choose $\{f_n^{K_j}\}_{n \in \mathbb{N}}$ to be a subsequence of $\{f_n^{K_{j-1}}\}_{n \in \mathbb{N}}$ for each $j > 1$, and consider the subsequence $\{f_n^{K_n}\}_{n \in \mathbb{N}}$. This sequence necessarily converges uniformly on each K_j because it is a subsequence of $\{f_n^{K_j}\}_{n \in \mathbb{N}}$ for each j , so we have shown that $\mathcal{F}$ being locally normal implies that it is normal; the contrapositive is what we wanted to prove. $\square$